Solution / Buns

2110201: COMP ENG MATH (2023/2) Quiz 1

Name: Solution Student ID: 0123456789 Entry Number: 0

1. (8 points) Given a function $f\colon \mathbf{R}^4 \Rightarrow \mathbf{R}^3$ such that

$$f(a,b,c,d) = (a+b+2d, 2a+3b-c+7d, b-c+3d)$$

Let $\mathbf{A}$ be a matrix that represents the linear transformation f.

1.1. (2 points) Find the matrix $\mathbf{A}$.

$$A = \begin{bmatrix} 1 & 1 & 0 & 2 \\ 2 & 3 & -1 & 7 \\ 0 & 1 & -1 & 3 \end{bmatrix}$$

1.2. (3 points) Find RRE of the matrix $\mathbf{A}$.

$$\begin{bmatrix} 1 & 1 & 0 & 2 \\ 2 & 3 & -1 & 7 \\ 0 & 1 & -1 & 3 \end{bmatrix}$$

$$\xrightarrow{R_2-2R_1} \begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & -1 & 3 \\ 0 & 1 & -1 & 3 \end{bmatrix}$$

$$\xrightarrow{R_3-R_2} \begin{bmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_1-R_2} \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\therefore RRE(A) = \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

1.3. (2 points) Find all vectors $\mathbf{x}$ such that $\mathbf{Ax} = \mathbf{0}$

$$\begin{array}{cccc} p & p & f & f \\ \end{array}$$
$$\begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 3 \end{bmatrix}$$
$$\begin{array}{cccc} 0 & 0 & 0 & 0 \end{array}$$

$$h_1 = \begin{pmatrix} p \\ p \\ f \\ f \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \end{pmatrix} \qquad h_2 = \begin{pmatrix} p \\ p \\ f \\ f \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \\ 0 \\ 1 \end{pmatrix}$$

2 answers:

$$x = \begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \end{pmatrix} s_1 + \begin{pmatrix} 1 \\ -3 \\ 0 \\ 1 \end{pmatrix} s_2, \quad s_1, s_2 \in \mathbb{R}$$

$$x \in \left\{ \begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \end{pmatrix} s_1 + \begin{pmatrix} 1 \\ -3 \\ 0 \\ 1 \end{pmatrix} s_2 \;\middle|\; s_1, s_2 \in \mathbb{R} \right\}$$

1.4. (0.5 points) Is the function f one-to-one? (Answer **Yes**/**No** only)

(one-to-one : every column has a pivot)

1.5. (0.5 points) Is the function f onto? (Answer **Yes**/**No** only)

(onto: every row has a pivot)

2. (2 points) Answer **True** or **False** for each sub question in the table provided.

2.1	2.2	2.3	2.4
False	True	True	True

2.1. $G(x,y,z) = (x^2, y)$ is a linear transformation.

2.2. Given matrix $\mathbf{T} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $\mathbf{x} = (1, 1)$, and $\mathbf{b} = \mathbf{Tx} = (3, 7)$. We can conclude that (1, 1) is a pre-image of (3, 7) respect to $\mathbf{T}$. $\rightarrow \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$ ✓

2.3. The vector (1, 1) is an eigenvector of the matrix $\begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$. $\rightarrow \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\therefore \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is eigenvector with $\lambda = 1$ *

2.4. Please answer **True** for a free point.